

Odontogenic Cysts and Tumors — Faculty Lecture

Oral Pathology Edu - Sample lecture for PDF to quiz generation

Odontogenic cysts are epithelial-lined cavities that arise from remnants of the tooth-forming apparatus. They are a central topic in oral pathology because many jaw radiolucencies are cystic in nature.

Radicular cysts develop at the apex of a non-vital tooth and are the most common inflammatory odontogenic cyst. They are associated with pulp necrosis and periapical inflammation.

Dentigerous cysts surround the crown of an unerupted or impacted tooth and are classified as developmental odontogenic cysts rather than inflammatory lesions.

The odontogenic keratocyst, also termed keratocystic odontogenic tumor in some classifications, may show aggressive growth and a higher recurrence rate after surgery than many other jaw cysts.

Ameloblastoma is the most frequent benign odontogenic epithelial tumor. It often presents in the posterior mandible as a slowly expansile multilocular radiolucency on panoramic radiographs.

Odontogenic myxoma can produce a soap-bubble or honeycomb radiographic pattern and may infiltrate between trabeculae, making complete surgical removal important for local control.

Periapical radiographs may show a continuous radiopaque lamina dura around healthy tooth roots. Loss or disruption of the lamina dura can suggest periapical inflammatory disease.

A well-defined unilocular radiolucency associated with an impacted third molar should raise suspicion for a dentigerous cyst until histopathology confirms the diagnosis.

Histologic examination remains the gold standard to distinguish ameloblastoma from other odontogenic tumors and from cystic lesions with overlapping imaging features.

Long-term follow-up after enucleation or marsupialization of keratocystic lesions is recommended because recurrence can occur years after initial treatment.

Clinical examination should include pulp vitality testing of adjacent teeth when evaluating a periapical radiolucency to help separate radicular cysts from non-inflammatory cysts.

Students should be able to explain why inflammatory radicular cysts differ in etiology from developmental dentigerous cysts despite similar radiographic appearances in some cases.